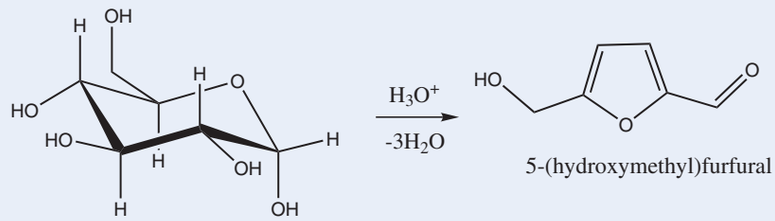
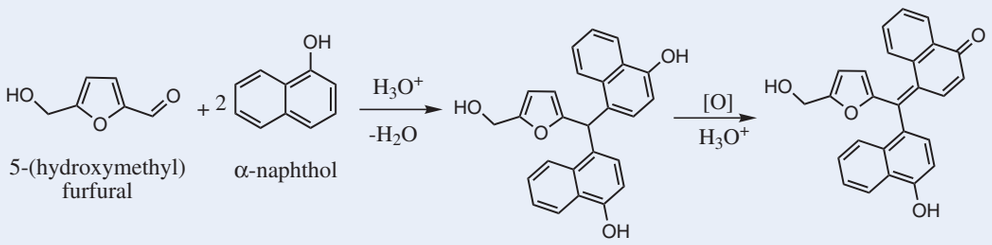
16. Molisch's test:

Disaccharides, and polysaccharidesare hydrolysed to Monosaccharides by strong mineral acids. Pentoses are then dehydrated to furfural, while hexoses are dehydrated to 5-hydroxymethylfurfural. These aldehydes formed will condense with two molecules of α-Naphthol to form a purple-coloured product, as shown below.



5-(hydroxymethyl)furfural



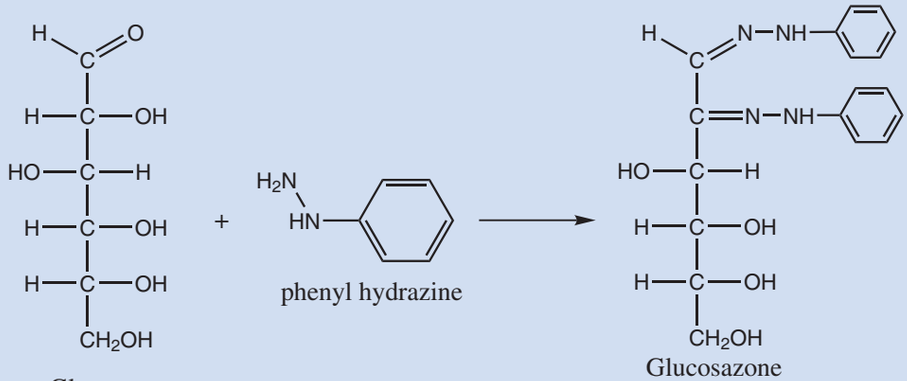
5-(hydroxymethyl) furfural

α-naphthol

A purple dye

17.Osazone test:

Phenyl hydrazine in acetic acid, when boiled with reducing sugars forms Osazone. The first two carbon atoms are involved in this reaction. The sugars that differ in their configuration on these carbon atoms give the same type of Osazone. Thus glucose, fructose and mannose give the same needle type yellow crystals.



Glucose

phenyl hydrazine

Glucosazone
(Yellow crystals)

organic analysis.indd 264 19-01-2021 12:24:22